

Pros and Cons of RFID Technology

Pros

1. RFID tags are rugged and robust and can work in harsh temperatures and environment. The RFID system works at a remarkably high speed, even in adverse conditions.
2. RFID tags are available in different shapes, sizes, types and materials. The information on the read-only tag cannot be altered or duplicated. Read-write tags can be used repeatedly. The RFID tags are always read without any error.
3. Direct physical contact between the tags and the reader is not required. RF technology is used for communication.
4. Multiple RFID tags can be read at the same time. The RFID tags can be read in a bulk of ten to 100 tags at a time. Reading of the tags is automatic and involves no labour.
5. RFID systems can identify and track unique items, unlike the bar code system which identifies only the manufacturer and the product type.
6. The entire RFID system is very reliable, which allows the use of RFID tags for security purpose.
7. The storage capacity of the RFID tags is greater than of any other automatic identification and tracking system.

Cons

1. The RFID system is costly compared to other automatic identification systems. The cost can increase further if the RFID system is designed for a specific application.
2. Size and weight of the tags is more as compared to barcode system. The electronic components like antenna, memory and other parts of the tags make them bulky.
3. Although the tags work in harsh environments, the signals from certain types of tags get affected when they come in close contact with certain metals or liquids. Reading such tags becomes difficult and sometimes the data read is erroneous.
4. There is no way in which damaged tags can be tracked and replaced with tags that are intact.
5. Although the tags do not require line-of-sight communication, they can be read within a specified range only.

pensive and bulky. Newer innovations in the RFID industry include active, semi-active and passive RFID tags. These tags can store up to 2 kilobytes of data and are composed of a microchip, antenna and, in the case of active and semi-passive tags, a battery. The tag's components are enclosed within plastic, silicon or sometimes glass. Table I gives the performance overview of the different-frequency passive tags.

Active and passive tags

The first basic choice when considering a tag is between passive, semi-passive and active. Passive tags can be read from a distance of up to 4-5 metres using the UHF frequency band, whilst the other types of tags (semi-passive and active) can achieve much greater communication distances of up to 100 metres for semi-passive and several kilometres for active. This large difference in communication performance can be explained by the following:

1. Passive tags use the reader field as a source of energy for the chip and for communication from and to the reader. The available power from the reader field not only reduces very rapidly with distance but is also controlled by strict regulations, resulting in a limited communication distance of 4-5 metres when using the UHF frequency

band (860-930 MHz).

2. Semi-passive (battery-assisted backscatter) tags have built-in batteries and therefore do not require energy from the reader field to power the chip. This allows them to function with much lower signal power levels, resulting in greater distances of up to 100 metres. Distance is limited mainly because the tag does not have an integrated transmitter, and still has to use the reader field to communicate back to the reader.

3. Active tags are battery-powered devices that have an active transmitter onboard. Unlike passive tags, these generate RF energy and apply it to the antenna. This autonomy from the reader means that they can communicate from distances of over several kilometres.

HF and UHF are best suited to the supply chain. UHF, due to its superior read range, will become the dominant frequency. LF and microwave will not be used in certain cases.

Tag ICs

RFID tag ICs are designed and manufactured using some of the most advanced and smallest-geometry silicon processes available. The result is impressive, when you consider that the size of a UHF tag chip is around 0.3 mm².

In terms of computational power, RFID tags are quite dumb, containing only basic logic and state machines capable of decoding simple instructions. This does not mean that they are simple to design. In fact, very real challenges exist such as achieving very low power consumption, managing noisy RF signals and keeping within strict emission regulations.

Other important circuits allow the chip to transfer power from the reader signal field, and convert it via a rectifier into a supply voltage. The chip clock is also normally extracted from the reader signal.

The amount of data stored on a tag depends on the chip specifications, and can range from just simple identifier numbers of around 96 bits to more information about the product containing up to 32 kbits. However, greater data capacity and storage (memory size) leads to larger chip sizes and hence more expensive tags.

In 1999, the AUTO-ID Center (now EPC Global) based at the Massachusetts Institute of Technology in the US, together with a number of leading companies, developed the idea of a unique electronic identifier code called the electronic product code (EPC). The EPC is similar in concept to the universal product code used in barcodes today.